

The Sector Crossing: What "Measurement" Actually Is

$\Sigma = 1$ Sector

(Coherent / Reversible)

- Photons (always)
- Superpositions (until crossing)
- Entangled pairs in transit
 - Unitary evolution
 - $Q < k_B T \ln 2$
- Erasure possible

SECTOR CROSSING
"Measurement"

$\mu < 1$ Sector

(Decohered / Irreversible)

- Matter (timelike worldlines)
 - Classical states
 - Measured particles
- Entropy: $\Delta S \geq k_B \ln 2$
 - $Q \geq k_B T \ln 2$
- Erasure impossible

THE LANDAUER CRITERION

Sector crossing occurs when: $Q_{\text{interaction}} \geq k_B T \ln 2$

$Q = \text{energy dissipated into environment by the interaction}$
 $T = \text{temperature of the environment}$

No observer required. No consciousness. Just thermodynamics.